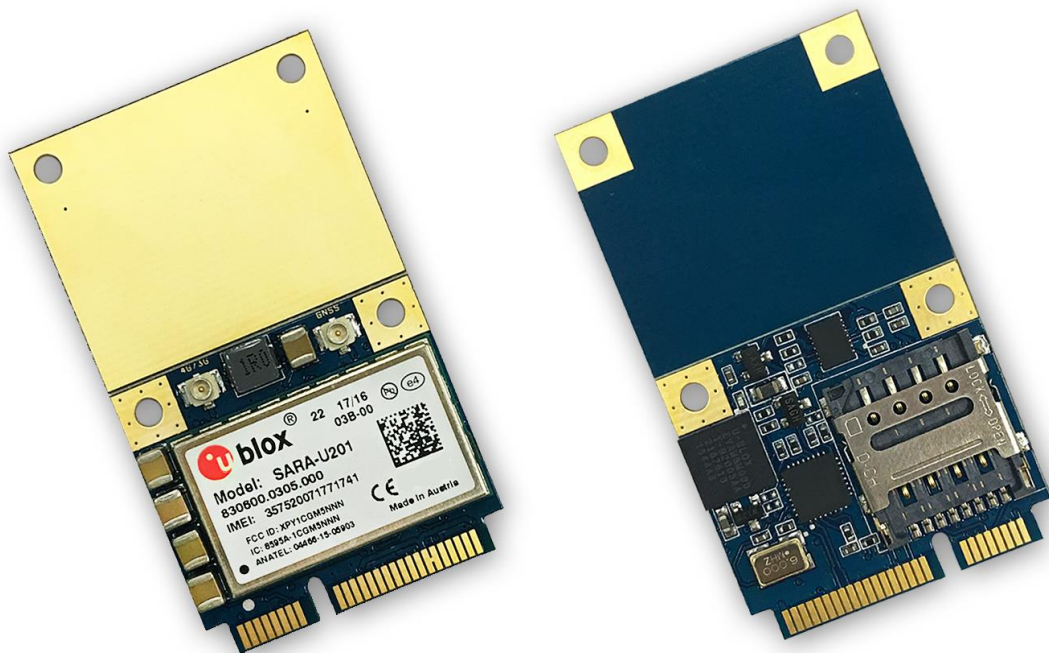


RXH2000 Series

3G WCDMA & GNSS Full / Half-size mini PCIe Card

Datasheet



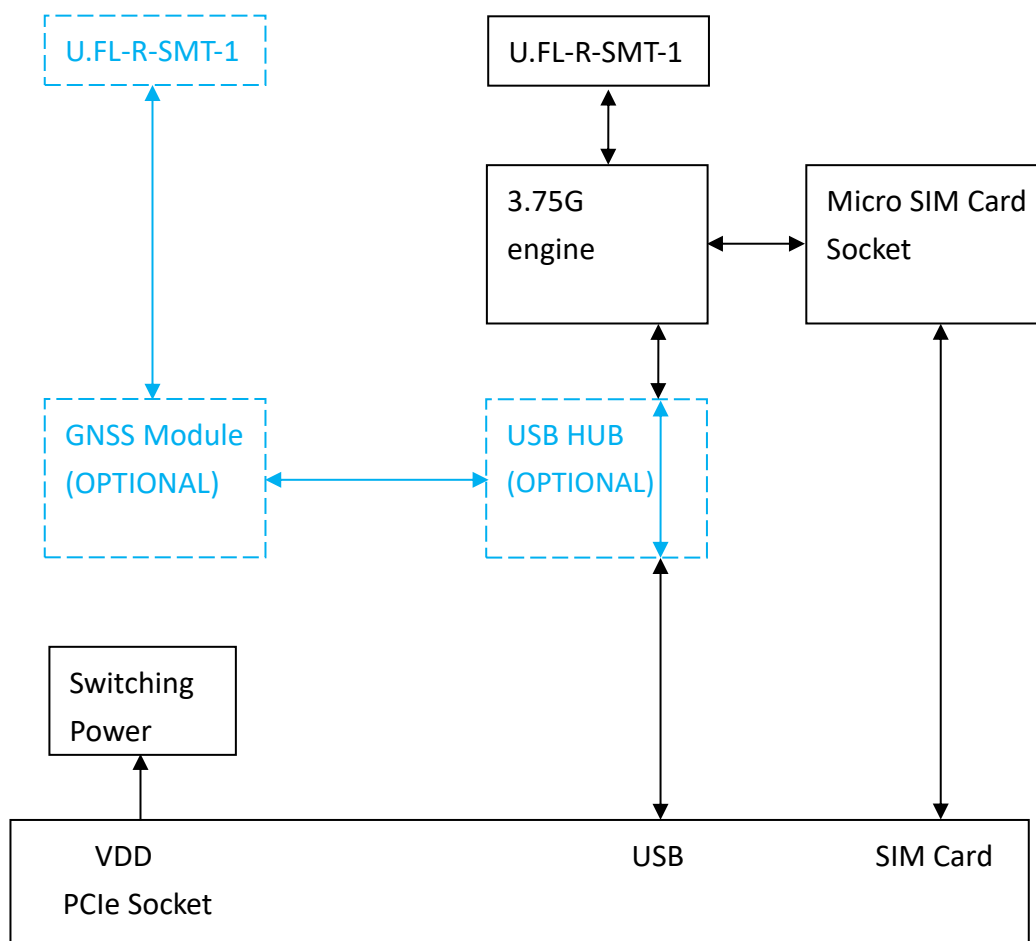
PRODUCT DESCRIPTION

The RYH2000 series is based on the PCI Express Mini Card standard with a USB 2.0 interface

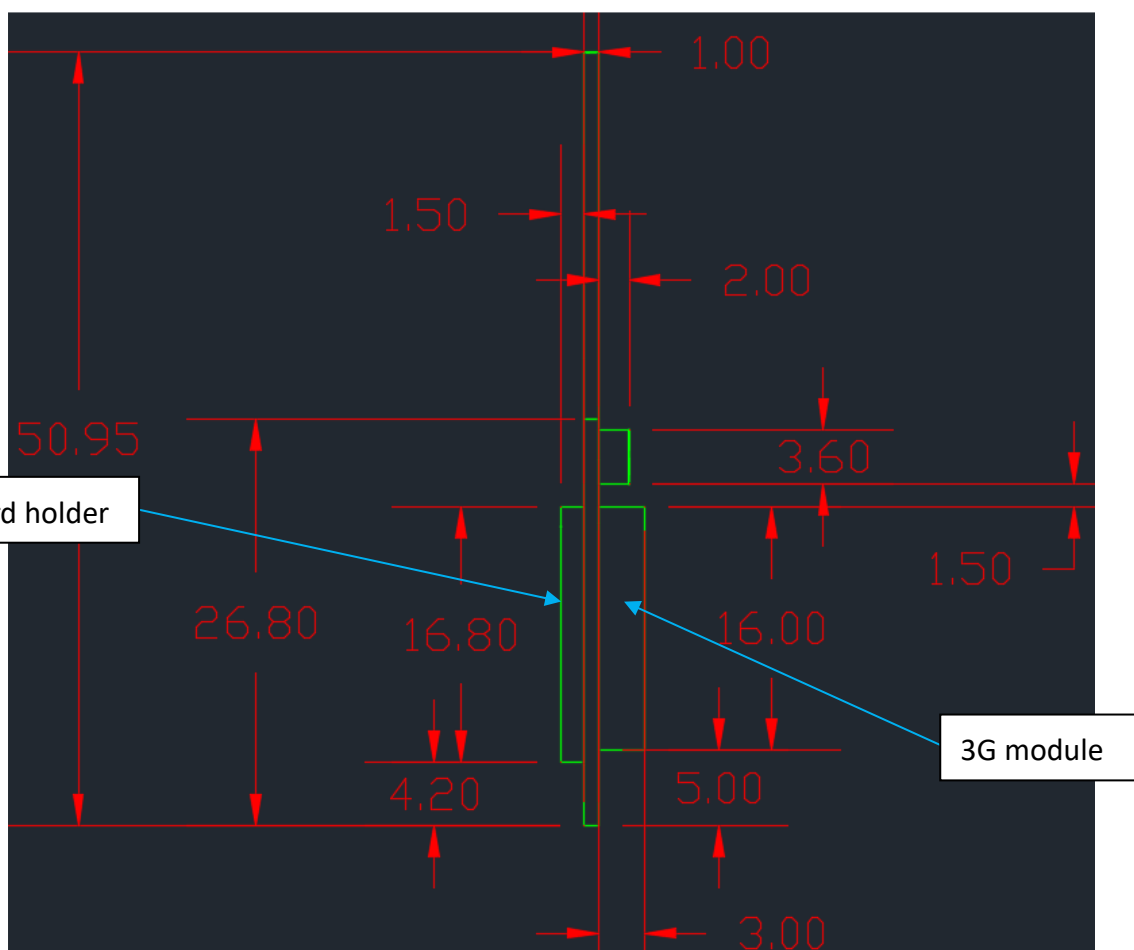
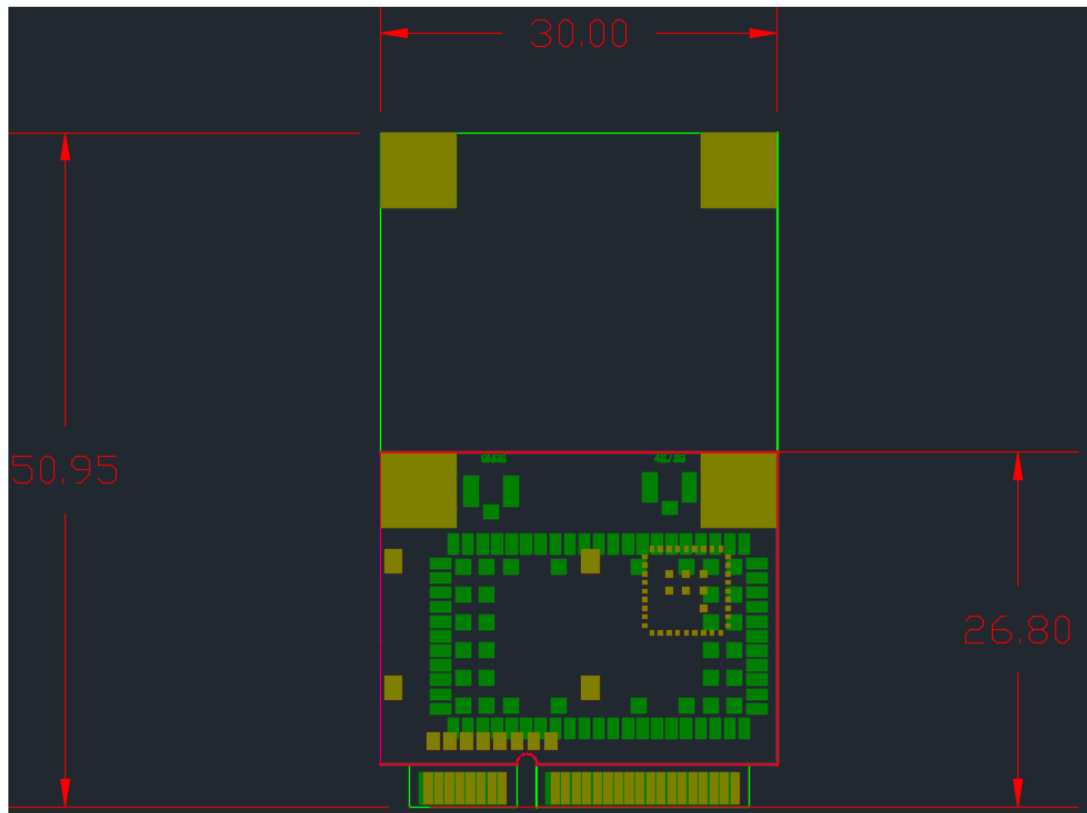
FEATURES

- u-blox SARA-U201, SARA-U260, SARA-U270 3.75G engine.
- UMTS/HSPA 800/850/900/1900/2100MHz. (OPTIONAL)
- Build-in a micro SIM card holder.
- u-blox 72-channel EVA-M8M-0 engine.(OPTIONAL)
- Standard mini PCIe full-size 30mm*50.95mm / half-size 30mm*26.8mm.
- Low idle mode current.
- Temperature range: -40 to +85°C.
- CellLocate®: location based on cellular networks.
- Manufactured in ISO/TS 16949 certified production sites.
- Mini PCIe Signal Type USB2.0.

BLOCK DIAGRAM



DIMENSIONS



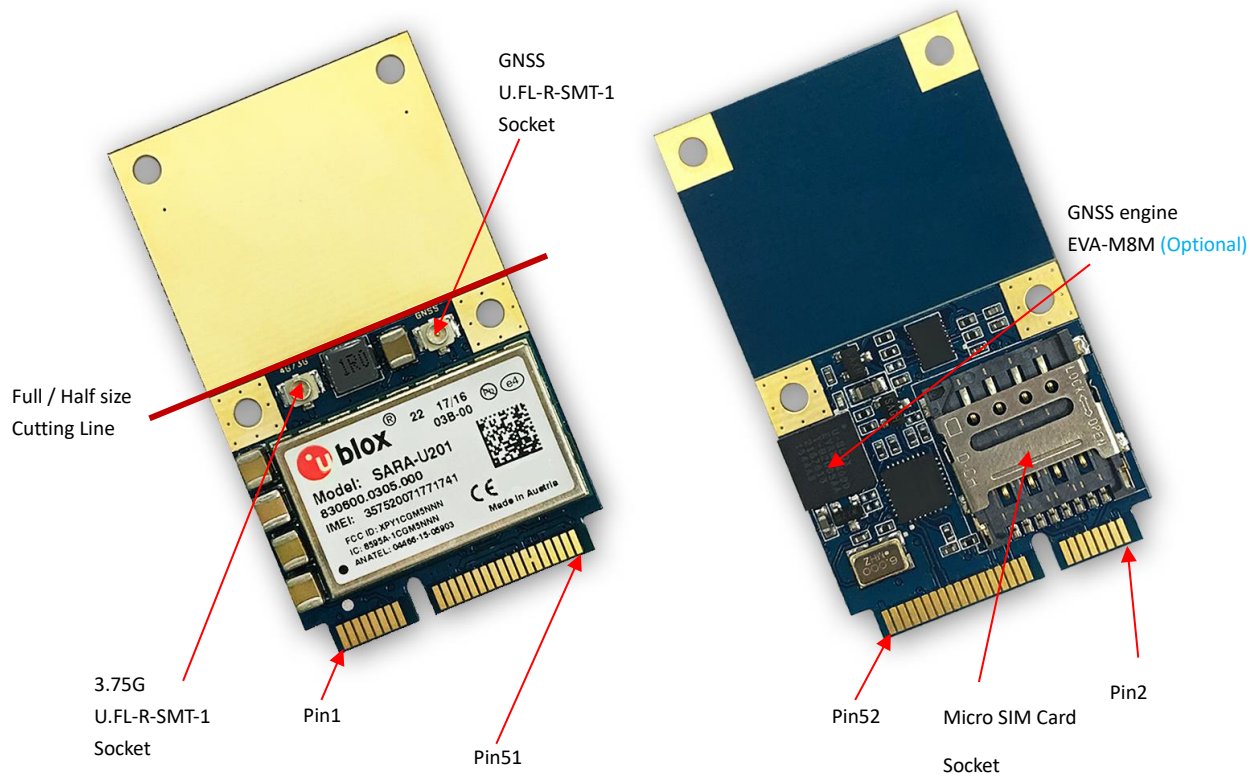
micro SIM card holder

3G module

unit: mm

PCB Thickness 1mm

PIN CONNECTOR



PIN DESCRIPTION

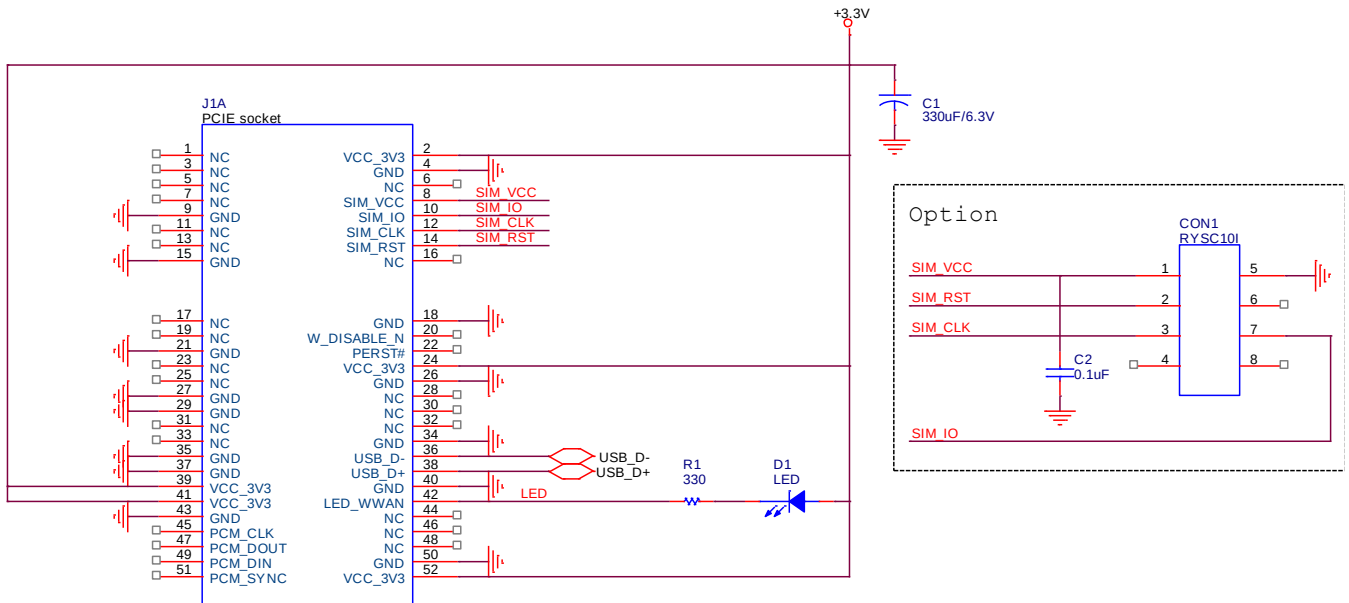
Pin	Name	Input/Output	Description
1	NC		Not connected
2	VCC_3V3	Power	Power Input
3	NC		Not connected
4	GND		Power Ground
5	NC		Not connected
6	NC		Not connected
7	NC		Not connected
8	SIM_VCC		External SIM signal – Power supply for the SIM
9	GND		Power Ground
10	SIM_IO	Input/Output	External SIM signal – Data I/O
11	NC		Not connected
12	SIM_CLK	Output	External SIM signal – Clock
13	NC		Not connected
14	SIM_RST	Input	External SIM signal – Reset
15	GND		Power Ground
16	NC		Not connected
17	NC		Not connected
18	GND		Power Ground
19	NC		Not connected
20	W_DISABLE_N	Input	If this pin is unused, keep open.
21	GND		Power Ground
22	NC		Not connected
23	NC		Not connected
24	VCC_3V3	Power	Power Input
25	NC		Not connected
26	GND		Power Ground
27	GND		Power Ground
28	NC		Not connected
29	GND		Power Ground
30	NC		Not connected
31	NC		Not connected
32	NC		Not connected
33	NC		Not connected
34	GND		Power Ground

35	GND		Power Ground
36	USB_DN	Input/Output	USB Data Negative
37	GND		Power Ground
38	USB_DP	Input/Output	USB Data Positive
39	VCC_3V3	Power	Power Input
40	GND		Power Ground
41	VCC_3V3	Power	Power Input
42	LED_WWAN	Output	Default: high impedance; LED setting flash via AT command port AT+UGPIOC=16,2,1
43	GND		Power Ground
44	NC		Not connected
45	NC		Not connected
46	NC		Not connected
47	NC		Not connected
48	NC		Not connected
49	NC		Not connected
50	GND		Power Ground
51	NC		Not connected
52	VCC_3V3	Power	Power Input

SPECIFICATION

Item	Min.	Typ	Max.	Unit	Description
u-blox 3G SARA-U201					
3.75G band		5		Ch.	800/850/900/1900/2100 MHz
GSM/GPRS/EDGE band		4		Ch.	850/900/1800/1900MHz
u-blox 3G SARA-U260					
3.75G band		2		Ch.	850/1900 MHz
GSM/GPRS/EDGE band		2		Ch.	850/1900 MHz
u-blox 3G SARA-U270					
3.75G band		2		Ch.	900/2100 MHz
GSM/GPRS/EDGE band		2		Ch.	900/1800 MHz
Downlink Speed		7.2		Mb/s	HSDPA
Uplink Speed		5.76		Mb/s	HSUPA
Operating Voltage Power Input VCC_3V3	3	3.3	5.5	V	
USB Interface					V2.0
Operating Temperature Range	-40	25	+85	°C	
Dimension					miniPCle Standard 26.8mm*30mm
Antenna Connector		1			U.FL-R-SMT-1
Weight		6		g	
u-blox GNSS EVA-M8M-0 (OPTIONAL)					
GNSS Center Frequency		1575.42 1602.5625		MHz	GPS Glonass
Navigation Update Rate		1	18	Hz	Configurable
Accuracy		2.5		M	CEP
Cold Starts		27		Sec.	
Aided Starts		4		Sec.	
Reacquisition		1		Sec.	
Tracking Sensitivity		-164		dBm	
Cold Starts Sensitivity		-147		dBm	
Hot Starts Sensitivity		-156		dBm	

APPLICATION CIRCUIT REFERENCE



DRIVER SUPPORT

- RIL software for Android
 - Embedded Windows 6.x, 7.x
 - Windows Mobile 6.5
- USB driver Embedded Windows 6.x, 7.x
 - Windows XP, Vista, 7, 8, 10
 - Windows Mobile 6.5
- Linux

Enable the kernel configuration as follows:

```
Device Drivers
  USB Support
    <*>USB Modem (CDC ACM) support
```

Enable the kernel configuration as follows:

```
Device Drivers
  USB Support
    <*>USB Modem (CDC ACM) support
```

Enable the kernel configuration as follows:

```
Device Drivers
  Network device support
    <*>PPP (point-to-point protocol) support
    <*>PPP support for async serial ports
    <*>PPP support for sync tty ports
    <*>PPP Deflate compression
```

DRIVER AND CONFIGURATION TOOLS

Complete software along with host driver for various operating systems such as Windows, Linux, Windows CE, and Android.

Please download the driver and u-blox u-center from <http://www.u-blox.com>

ORDER INFORMATION

Ordering No.	3G Engine	3.75G Band	GNSS Engine
RYH201I	SARA-U201	800/850/900/1900/ 2100 MHz	
RYH2018	SARA-U201	800/850/900/1900/ 2100 MHz	EVA-M8M-0
RYH270I	SARA-U270	900/2100 MHz	
RYH2708	SARA-U270	900/2100 MHz	EVA-M8M-0